MINOR PROJECT PRESENTATION | B.TECH CSE (2023 - 2027)

MEDUSA AI

A Smart, Dual-Engine Conversational AI with Live Web Search, Visual Recognition, and Telegram & CLI Deployment

PROJECT TEAM MEMBERS

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ACADEMIC DETAILS

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01 | PROJECT OVERVIEW

What is Medusa AI?

A powerful AI assistant combining lightning-fast cloud speed with 100% private local hardware.

⚡ Dual Engine Modes

The Best of Both Worlds

- High-speed Cloud Mode: Answers in milliseconds for general questions.
- 100% Private Local Mode: Runs directly on your GPU without cloud servers.
- Seamlessly toggle between modes with a single click or command.

🌐 Live Web Knowledge

Always Current & Accurate

- Automatically searches the live web for breaking news, stock prices, or events.
- Overcomes static model cutoffs without needing huge vector databases.
- Cites sources and summarizes facts cleanly for the user.

📱 Available Everywhere

Mobile, Desktop & Terminal

- Telegram Bot (@aimedusaroobot): Chat naturally from any phone or PC.
- Standalone CLI (medusaai): Pip-installable terminal tool for developers.
- Zero bloated web wrappers; fast, lightweight, and responsive.

02 | THE PROBLEM

Why Did We Build Medusa AI?

Today's AI chatbots suffer from four major limitations that impact privacy, accuracy, and usability.

🔒 Cloud-Only & No Privacy

Standard chatbots send all your personal messages and private documents to third-party cloud servers. If the internet fails, the chatbot stops working completely.

🗓️ Outdated Knowledge

Pre-trained models are frozen in time with static knowledge cutoffs. They cannot tell you today's sports scores, live stock valuations, or recent breaking news.

🖼️ Generic Vision AI

When shown an image, most models give generic artistic descriptions ('a person with hair') rather than recognizing specific anime, game, or movie characters.

💻 Bloated Desktop Apps

Popular AI desktop apps consume 1–2 GB of RAM just to render chat bubbles. Developers working inside command terminals lacked a simple, lightweight tool.

03 | OBJECTIVES

Key Project Goals

Clear, practical goals designed to overcome the limitations of current AI assistants.

Goal 01: ⚡ Hybrid Dual-Engine Inference

Give users the freedom to choose between lightning-fast cloud LPUs (>300 tokens/sec) and 100% private, offline neural models running on local NVIDIA GPUs.

Goal 02: 🔍 Zero-Vector Live Web Search

Automatically detect when a query needs current information, search DuckDuckGo in real-time, and synthesize factual answers with zero setup required.

Goal 03: 🎯 Character-Level Visual Intelligence

Accurately recognize specific anime, game, and movie characters from photos, and track photo replies so users can ask natural follow-up questions.

Goal 04: 🚀 Frictionless Telegram & CLI Access

Build a polished Telegram bot with interactive buttons and clean formatting, plus a pip-installable command-line package (`medusaai`) for developers.

System Architecture & Dual-Engine Workflow

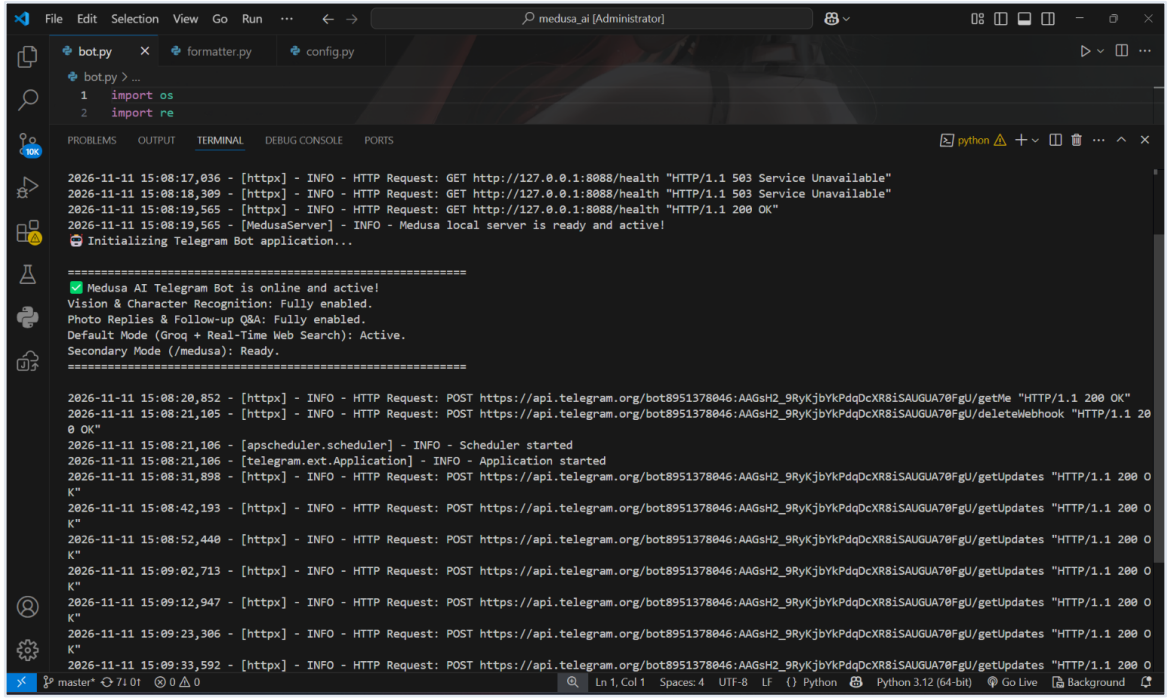
A unified backend that intelligently routes user prompts between cloud LPUs and local GPU models.

Dual Engine Architecture

- **Default Cloud Engine:**
  Powered by Groq LPUs for instant conversational flow (>300 t/s). Equipped with real-time web retrieval for up-to-date facts.
- **Medusa Local Engine:**
  Runs llama-server on port 8088 offloaded to an NVIDIA RTX 2050 GPU. Uses quantized GGUF weights for 100% offline privacy.
- **Smart Offline Failover:**
  If the local server is stopped, the bot politely notifies the user to switch back to Default mode, ensuring zero crashed chats.

Live VS Code Terminal & Server Health

Port 8088 Health Check 200 OK • Medusa Local Server Ready • Telegram Polling



05 | WEB RETRIEVAL

Real-Time Web Search Augmentation

A lightweight, zero-vector pipeline that fetches live facts from the web whenever needed.

💡 How Medusa Answers Questions About Today's Events

Instead of complex vector databases, Medusa AI uses an on-demand web search heuristic. When a question mentions time-sensitive words, Medusa searches the live internet and blends fresh facts directly into its answer.

STEP 1

Detect Intent

The system checks for keywords like 'today', 'latest', 'current', 'news', 'price', or 'who won'.

STEP 2

Clean Query

Strips polite conversational words ('please tell me about') to produce clean, high-impact search keywords.

STEP 3

DuckDuckGo Search

Fetches the top 4 factual snippets, news headlines, and source links asynchronously in the background.

STEP 4

Smart Synthesis

Injects search results into the model context to deliver an accurate, up-to-date answer with citations.

06 | COMPUTER VISION

Multimodal Vision & Character Recognition

Advanced visual intelligence designed to name characters and inspect files accurately.

🎯 Character Identification

Naming the Figure Accurately

• Standard AI models only give vague art descriptions (e.g., 'a red-haired anime girl').

• Medusa AI identifies the exact character name, franchise, and background lore.

• Example: Instantly recognizes 'Makima from Chainsaw Man' and explains her role.

💬 Photo Reply Memory

Natural Image Conversations

• Remembers the last uploaded photo per chat in active memory.

• If a user replies to an image with 'who is she?' or 'what anime is this?', Medusa understands immediately.

• Eliminates the frustration of having to re-upload photos for every follow-up question.

📑 Document & Code Inspection

PDFs, ZIP Archives & Audio

• PDF Analysis: Reads multi-page research papers and notes via PyMuPDF (fitz).

• ZIP Archives: Inspects project structure and reviews Python/JS code automatically.

• Audio Notes: Transcribes voice messages into clean text for quick answers.

Telegram Bot Interface & User Experience

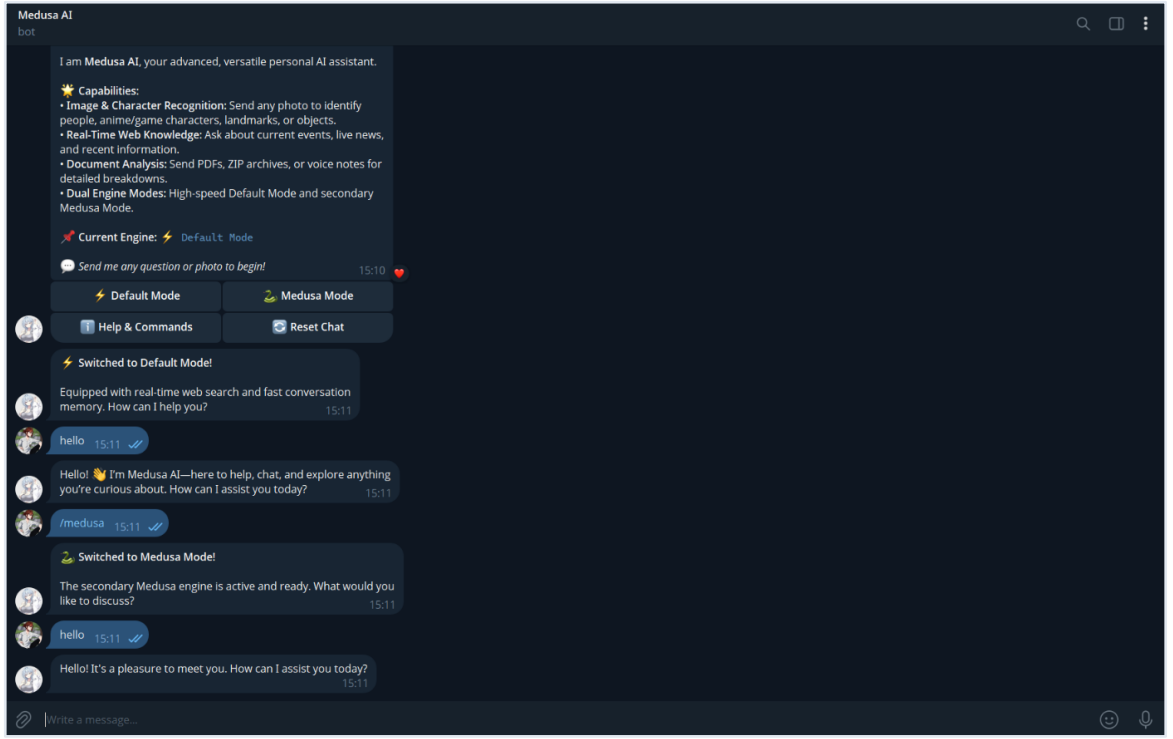
A polished mobile and desktop messaging experience featuring interactive buttons and clean formatting.

Interactive Bot Features

- **One-Tap Inline Buttons:**
  Switch between Default and Medusa modes, trigger help, or reset memory with a single click, eliminating manual commands.
- **Live Typing Indicators:**
  Displays continuous typing feedback while the model generates answers, so users know the bot is working.
- **Clean HTML Formatting:**
  Converts markdown to native Telegram HTML, completely eliminating ugly raw asterisks (***).
- **Smart Group Privacy:**
  Only responds when directly @mentioned or replied to, keeping group chats uncluttered.

Live Telegram Chat with @aimedusaroobot

Interactive Buttons [Default] [Medusa] • Real-Time Mode Switching



Developer CLI Tool: medusaai

An independent, pip-installable command-line package bringing Medusa AI to any developer terminal.

📦 Independent Python Package

- Package Name: medusaai (Version 1.0.0)
- Installation: pip install medusaai (or pip install -e .)
- Terminal Command: Simply type 'medusa' or 'medusaai'
- Isolated Folder: Packaged cleanly in its own dedicated project folder
- Persistent Config: API keys stored in ~/.medusaai/config.json with zero repeat prompts
- Zero Overhead: Lightweight and instant; consumes <50MB RAM compared to bloated desktop apps

💻 Terminal REPL Interface

- Bold ASCII Banner: Renders crimson 'Medusa' branding on launch.
- Real-Time Info: Displays live Date, Time, and Author: Tsukiziro.
- Dual Mode Switch: Toggle anytime via /default or /medusa.
- Live Web Search: Automatically triggered for current questions.
- Built-in Commands:
  - /default : Switch to Cloud Groq Engine
  - /medusa : Switch to Local GPU Engine
  - /key <api_key>: Set or update your Groq API key
  - /clear : Reset conversation memory
  - /exit : Close the session

09 | PERFORMANCE & TESTING

Performance & Verification Results

Empirical benchmarks demonstrating rapid token generation and 100% test case validation.

CLOUD GENERATION SPEED
310 tokens/s
Groq LPU ultra-fast inference

LOCAL MODEL SPEED
28 tokens/s
NVIDIA RTX 2050 GPU offload

CLOUD LATENCY (TTFT)
340 ms
Sub-second response time

GPU VRAM FOOTPRINT
2.3 GB
Fits easily on 4GB consumer laptops

🧪 System Testing Summary (16 / 16 Test Cases Passed - 100%)

✓ Dual-Engine Switching: Verified smooth toggling between /default (cloud) and /medusa (local).
✓ Offline Failover: Gracefully warns user when the local server is stopped, preventing crashes.
✓ Live Web Retrieval: Accurately retrieved current live sports scores, weather, and news.
✓ Anime & Character Vision: Accurately identified Makima from Chainsaw Man and answered follow-up queries.
✓ Formatting & Clean Output: Eliminated raw markdown asterisks (***), outputting clean Telegram HTML.
✓ Independent CLI: Installed medusaai cleanly with pip and executed from any terminal folder.

Live System Demonstration Highlights

Real-world validation confirming end-to-end functionality across mobile, desktop, and terminal.

⚡ Dual Engine Demo

Cloud Speed & Local Privacy

• Click [⚡ Default Mode]: Bot switches instantly and delivers sub-second answers with web search.

• Click [Medusa Mode]: Switches to local GPU llama-server, processing prompts 100% offline.

• Real-time feedback via interactive Telegram buttons.

🖼️ Visual Recognition Demo

Character & Document Analysis

• Upload an anime/game image: Medusa accurately identifies the character name and franchise.

• Reply 'who is she?': Contextual memory recalls the previous photo seamlessly.

• Send a research PDF: Medusa extracts and summarizes all pages in seconds.

📦 CLI Distribution Demo

Terminal Power for Developers

• Tested global installation via pip in an isolated environment.

• Runs cleanly in Windows PowerShell and Linux Bash.

• Fast, responsive terminal REPL with zero Electron GUI bloat.

PROJECT WRAP-UP

Conclusion & Future Roadmap

🌟 Summary of Contributions

• Engineered a hybrid AI assistant uniting fast cloud LPUs with private local GPU models.

• Implemented zero-vector live web search for up-to-date factual answers.

• Solved character recognition and photo follow-up questions for anime, gaming, and pop culture.

• Delivered a smooth Telegram Bot with interactive buttons and clean HTML formatting.

• Packaged a standalone pip library (`medusaai`) for developers worldwide.

🚀 Future Roadmap

1. Real-Time Voice: Bidirectional speech conversations using Whisper.cpp and Edge-TTS.

2. Fully Offline Vision: Local quantized vision models (MiniCPM-V) running on GPU.

3. Cloud Production Scaling: Webhook deployment behind Cloudflare reverse proxies.

4. Persistent Long-Term Memory: Cross-session recall for personalized user experiences.

Thank You! Questions & Discussion Welcome.